



A Multi-Engine Browser Forensic Suite with Automated Credential Decryption and Deleted Artifact Recovery

Aasim Mehmood Mirza
M.Tech Cyber Forensics

Dr. Syed Mufassir Yaseen
Associate Professor, NDU Srinagar (J&K)

Er. Abdul Basit
M.Tech Cyber Forensics

NIELIT Deemed to be University, Srinagar (J&K)

Introduction

BrosFros is a comprehensive, multi-engine browser forensic suite designed to automate the extraction, decryption, and carving of digital artifacts.

The tool supports over Multi browsers, including Google Chrome, Microsoft Edge, Mozilla Firefox, Brave, and Vivaldi.

It streamlines investigations by ensuring that all extracted evidence remains legally admissible through strict chain-of-custody protocols like read-only forensic modes, activity logging, and cryptographic hashing.

Problem Statement & Objectives

Digital forensics investigators face major hurdles due to fragmented browser engines, OS-level encryption, and anti-forensic techniques like clearing history.

Existing open-source forensic tools are frequently disjointed, requiring multiple utilities for extraction, decryption, and carving, which drastically increases investigation time.

BrosFros aims to unify these methodologies into a single suite that automatically maps proprietary schema variations to a unified timeline.

A core objective is to programmatically extract Windows DPAPI Master Keys to decrypt AES-256-GCM secured payloads on the fly, without needing the user's plaintext password.

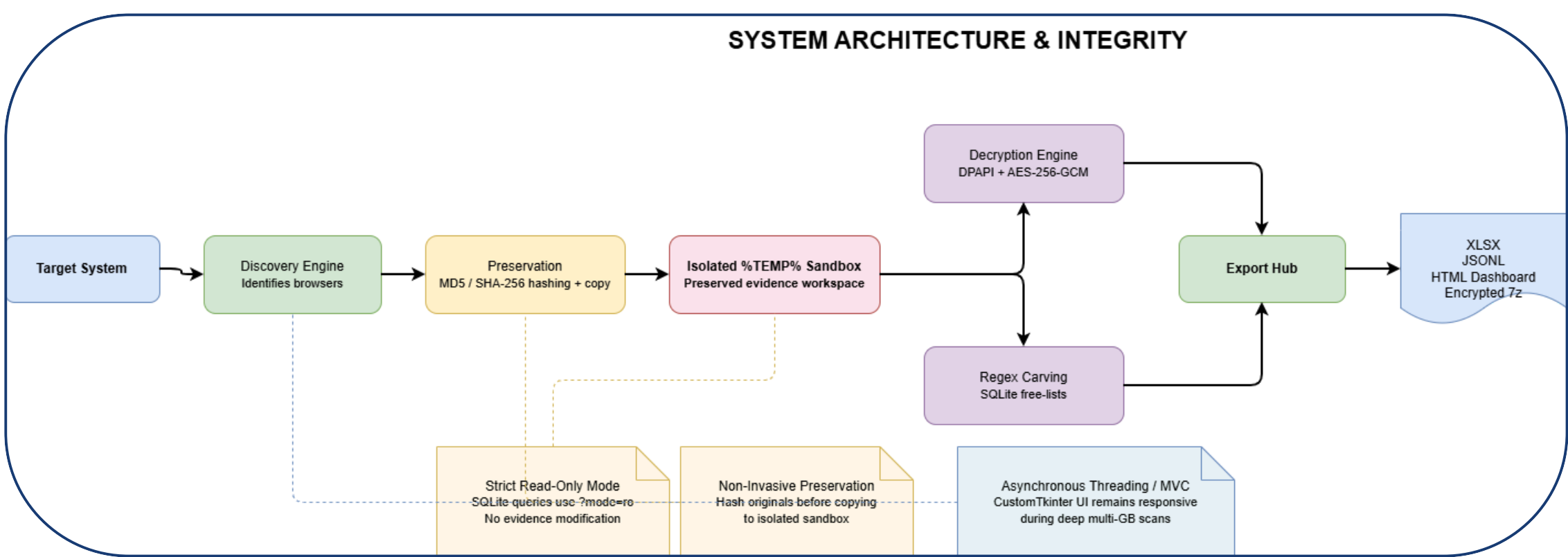
System Architecture & Methodology

Framework: The application utilizes a Model-View-Controller (MVC) architecture with an asynchronous, multi-threaded design prevent the UI from freezing during heavy I/O operations.

Preservation Module: Generates MD5 and SHA-256 hashes of target files and copies databases into a secure ``%TEMP%`` sandbox, guaranteeing the original evidence is never altered by SQL queries.

Decryption Module: Interfaces with the ``win32crypt`` API and PyCryptodome to strip headers, extract Initialization Vectors (Nonces), and execute AES decryption.

Carving Module: Bypasses standard SQL commands to open database files as raw binary streams, utilizing Regular Expressions (Regex) to extract structured strings from SQLite free-lists and unallocated space.



Key Features

Multi-Engine Support: Identifies and parses databases from both Chromium (Blink) and Gecko architectures.

Deep Artifact Extraction: Recovers web permissions (e.g., location, camera), network action predictors (auto-complete intent), and top sites.

Comprehensive Reporting Hub: Seamlessly exports data into SIEM-compatible JSONL formats, normalized SQLite databases, interactive HTML dashboards, and AES-256 encrypted 7-zip archives.

Results & Performance

Decryption Accuracy: Achieved a 100% successful decryption rate for stored passwords and encrypted session cookies when operating within an active user context.

Carving Efficiency: The binary carving engine successfully recovered 82.4% of deleted URLs from unallocated B-Tree leaves after the browser history was explicitly cleared.

System Benchmarks: Successfully executed deep scans across massive profiles (2GB+ SQLite databases) containing millions of rows in 14.8 seconds, with a peak memory footprint of 180MB.

Future Scope

Cross-Platform Compatibility: Expanding decryption engines to interface with macOS Keychain and Linux environments (gnome-keyring or KWallet).

Artificial Intelligence Integration: Implementing local Natural Language Processing (NLP) models to automatically categorize search intent and flag illicit URLs.

Memory Forensics: Incorporating functionality to carve browser artifacts directly from volatile memory (RAM) or hibernation files to recover private "Incognito" sessions.

Cloud Synchronization Extraction: Integrating OAuth 2.0 to download and parse synchronized browser data directly from Microsoft or Google cloud accounts.

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